Supplementary Information

Allosteric Protein Chemical Shift Perturbations are Ubiquitous

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Supplementary Text

The CSPdb

The CSPdb includes receptors that regulate diverse biological processes, including cell cycle progression (e.g., MAD2: 1KLQ), transcriptional control (e.g., Nrd1: 2MOW, 6GC3), translational regulation, autophagy, apoptosis (e.g., calcineurin: 2JOG; BCL-XL family: 2M04, 2LP8), and cell adhesion (e.g., tensin: 2LOZ). Disease-relevant targets address mechanisms in leukemia and other cancers (e.g., p53: 2MEJ), Alzheimer's disease (e.g., amyloid precursor protein-binding proteins: 2ROZ), and infectious diseases from viruses and bacteria, such as anthrax (2RUI), chikungunya (5I22), SARS-CoV-2 (7PKU), HIV (2L6E), and Ebola (2KQ0).

The CSPdb features several recurring protein domains that exhibit notable conformational changes upon ligand binding. For instance, it includes 23 interactions involving calmodulin (CaM) or CaM-like proteins with peptides (e.g., PDB IDs: 1SY9, 2JZI; full list in **SI Table S1**), which often involve substantial rearrangements upon binding described in detail elsewhere(1). Other receptors known for binding-induced conformational shifts or folding include the G protein Cdc42 (1CF4), the metaphase chromosome spindle checkpoint protein MAD2 (1KLQ), the KIX domain (2LXS), and CREB-binding proteins (e.g., 1R8U, 1L8C).

Additional recurring domains include the extra-terminal (ET) domains of bromodomain proteins BRD3 and BRD4 (e.g., 2NCZ, 2ND0; full list in **SI Table S7**, **SI Figure 6**), which bind intrinsically-disordered regions of viral integrase or chromatin regulator proteins. These interactions trigger subtle allosteric changes distal from the binding site, including the formation of a stabilizing secondary structure that reorients two allosteric helices (2, 3). Similarly, the PDZ2 domain (e.g., 1D5G, 1VJ6, 2M0V) shows allosteric effects mediated by the β2–β3 loop during peptide recognition (4).

Nine entries involve the PH domain of the TFIIH transcription initiation factor subunit Tfb1 (e.g., 2LOX, 2M14; full list in **SI Table S8**, **SI Figure 7**). In these complexes, the ligand is often modeled as a helical polypeptide bound orthogonally to an extended β-sheet, resulting in propagation of allosteric signals via hydrogen-bond rearrangements. Other targets which involve hydrogen bond rearrangement in β-sheets include the Cdc42 (1CF4), Cyclophilin A (2MS4), and SH3 domain (e.g., 2OJ2, 2ROL; full list in **SI Table S1**). An additional eight entries use ubiquitin or ubiquitin-like (e.g. SUMO and NEDD8) protein receptors. Ubiquitin has been used as a model system for NMR studies (5) and for studying allostery (6) (e.g., 2KWV, 2MUR; full list in **SI Table S9**, **SI Figure 8**).

In Other Supplementary Information, you will find a All_Case_Studies.pdf, which provides a concise summary for each of the 139 targets in the CSPdb. Each case study contains the following panels: A) a $^{15}N/^{1}H$ HSQC scatterplot overlaying the apo and holo assigned spectra with the optimal N/H offset applied; B) Calculated CSPs for receptor residues along the sequence, bars colored by confusion matrix classification; C) the mask of significant CSPs; D) the mask of binding-site residues; E) the confusion matrix classification projected onto the structure: allosteric CSPs are purple (FP), CSPs in the binding site are green (TP), non-CSPs in the binding site are orange (FN), and non-CSPs outside of the binding site are blue (TN). Residues without CSP measurements are shown in gray; ligands are colored cyan.

SI Table S1. Data for all receptors in this study

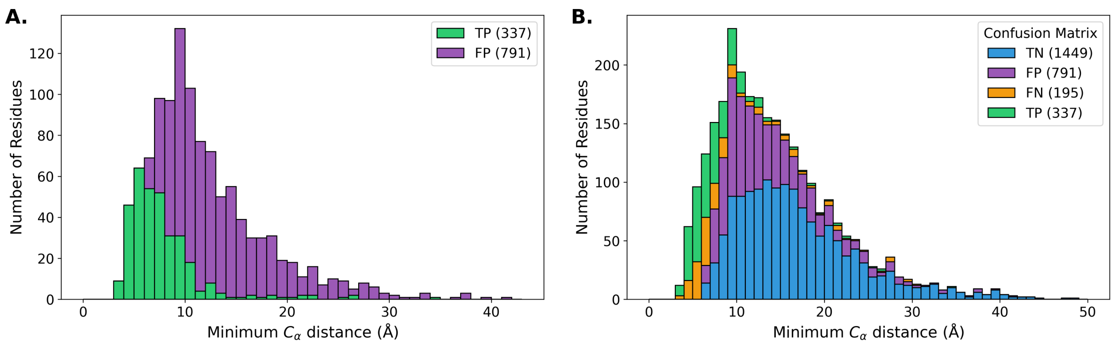
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	1YWI	6658	6659	0.78	0.72
	2FIN	5073	7024	0.65	0.52
	2ROZ	10235	10236	0.59	0.50
	5I22	4871	30010	0.55	0.49
	2LBM	15001	17569	0.60	0.46
7JMY	7JYN	30782	30790	0.53	0.43
	2N8T	25866	25865	0.64	0.42
	2C52	15398	6874	0.56	0.41
	2PLD	5318	5310	0.46	0.40
7JMY	7JYZ	30782	30791	0.50	0.40
	2L6E	16555	17307	0.52	0.40
	2RR4	11361	11115	0.56	0.40
	5MF9	34068	34067	0.54	0.39
1W1F	1WA7	6261	6456	0.51	0.38
	2L5E	50148	17270	0.38	0.37
2LSY	2N1G	18455	25559	0.45	0.37
7JMY	7JQ8	30782	30786	0.51	0.37
	6G04	34247	34248	0.43	0.37
2JNS	6BNH	15125	30373	0.48	0.36
	2LI5	16835	17879	0.42	0.36
	9QLM	15670	34986	0.59	0.34
	2ROL	15710	11040	0.43	0.34
	6FDT	19757	34224	0.33	0.34
	2M5A	5656	19043	0.53	0.34
	2KZU	17018	17019	0.38	0.34
	2MC6	19428	19429	0.47	0.33
	1M4P	50765	5532	0.29	0.32
	6K5T	25299	36260	0.46	0.32
2LSY	2LSK	18455	18434	0.38	0.32
	2MUR	17769	25230	0.40	0.32
	2M0U	18824	18825	0.35	0.31
	6OQJ	30599	30605	0.47	0.31
	2LVO	18581	18582	0.50	0.30
	2N4Q	18094	25677	0.40	0.30
	2KJ4	4276	16311	0.38	0.30
	6E5N	25544	30500	0.40	0.30
	2LY4	11532	18709	0.40	0.30
	7KLR	30808	30809	0.55	0.29
	2OJ2	4122	6581	0.37	0.28
	6GC3	17173	34260	0.29	0.28
	6K5R	6801	36259	0.43	0.28
	5VF0	17769	30276	0.31	0.27
	2LOZ	10318	17364	0.38	0.27
	2K17	15670	15671	0.53	0.27
	2KC8	16066	16065	0.52	0.27
	2L29	17128	17127	0.26	0.26
	2LP0	25142	17407	0.49	0.26
	2MUR	25229	25230	0.57	0.26
	2K6Q	5958	15877	0.36	0.26
	7X5C	51441	36473	0.31	0.26
	2LXM	18681	18682	0.30	0.25
2JNS	2NCZ	15125	26041	0.40	0.25
	2LGK	17813	17812	0.52	0.25
	1H8B	17627	4453	0.33	0.25
	7PKU	50446	34661	0.42	0.25
	6R5G	28070	34384	0.39	0.25
	7S5J	30926	30950	0.39	0.25
	2K7L	27288	15919	0.39	0.25
1U2N	1R8U	6268	5987	0.56	0.24
	2KYL	15972	16967	0.41	0.23
	2L4T	17254	17255	0.31	0.23
	8SG2	27579	31080	0.44	0.23
	2M3M	17373	17942	0.42	0.22
	1DDM	4683	4263	0.22	0.22
1SSF	2LVM	5878	18579	0.31	0.20
	2M56	17415	19038	0.12	0.18
	2B0F	5659	6823	0.27	0.18
	2YS5	11095	11094	0.35	0.18
6VED	2L3R	30703	17200	0.30	0.18
	5J8H	51289	30063	0.43	0.18
	5GOW	4901	36013	0.39	0.18
1SSF	2MWP	5878	25348	0.25	0.17
	6H8C	18827	34307	0.29	0.17
	2KVQ	15490	16788	0.25	0.16
	2L1R	30966	17103	0.35	0.16
	2MV7	18094	19516	0.38	0.15
	2KWI	15524	15525	0.40	0.15
	1VJ6	5131	6060	0.29	0.15
	2LP8	18250	18238	0.18	0.15
	2MZW	17503	25504	0.29	0.14
	7ZEY	16983	34727	0.44	0.14
	2M0K	51289	17771	0.41	0.14
	2MRE	18610	25070	0.28	0.13
	2KTB	51472	16694	0.20	0.13
	5JYV	30092	30093	0.31	0.13
	1PD7	6899	5808	0.37	0.13
1V49	2N9X	5958	25919	0.28	0.13
	2NMB	4683	4263	0.15	0.13
	2JW1	15503	15504	0.30	0.13
1SSF	2MWO	5878	25347	0.26	0.12
	2MES	51289	19238	0.45	0.11
2KQ3	2KHS	16585	15357	0.31	0.11
	2JQR	4566	15301	0.24	0.11
	2H7D	15792	7150	0.24	0.11
	2LE8	16396	17702	0.21	0.10
	2KOH	27205	16507	0.30	0.10
	2RSE	50325	11471	0.10	0.10
	2FIN	6809	7024	0.10	0.09
2JNS	2ND0	15125	26042	0.31	0.09
	2HUG	6592	7241	0.35	0.09
	2LTO	10276	18490	0.14	0.09
	2K7A	5461	16809	0.21	0.09
	2L0Y	15763	17067	0.27	0.09
	2LZ6	18610	18737	0.29	0.08
	2N55	52209	25694	0.44	0.07
	2MJV	15057	19738	0.30	0.07
	2RUI	16811	11570	0.16	0.06
	1SY9	51289	5480	0.36	0.06
	2KFT	6374	16878	0.52	0.06
	2RSY	7141	11508	0.30	0.05
	2M56	4154	19038	0.12	0.04
	2G35	15792	7061	0.17	0.04
	2LSP	15057	18439	0.26	0.03
	2M0J	51289	17771	0.35	0.03
2L5U	2L75	17285	17344	0.44	0.02
	7SFT	36097	30958	0.27	0.02
	2MNZ	19913	19914	0.44	0.02
	5U5S	50149	30206	0.19	0.01
	2MSR	34179	25130	0.12	0.00
	2KVM	15674	16778	0.33	-0.00
	2N80	25883	25829	0.26	-0.02
	2MWN	15792	25343	0.17	-0.03
	2LGG	17813	17808	0.37	-0.03
	2I94	5332	7293	0.17	-0.03
	1NPQ	4232	5738	0.24	-0.03
	2K8F	34231	15944	0.22	-0.05
2JNS	2ND1	15125	26043	0.23	-0.06
	5OEO	51289	34161	0.16	-0.07

(continued)

apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2JZI	51289	15624	0.43	0.22	2JNS	6BGG	15125	30367	0.14	-0.08
	7ZEY	34724	34727	0.36	0.22		6U4N	17827	30659	0.10	-0.08
	2CZY	27227	6749	0.38	0.22		2LGF	51289	17807	0.28	-0.09
7QCX	1D5G	34688	4516	0.27	0.22		2N83	25883	25833	0.29	-0.10
	2M5B	50942	19045	0.30	0.22		2JMF	6262	15016	0.13	-0.17
	1KLQ	52275	5299	0.32	0.22		2K6D	52080	15866	0.15	-0.18

SI Table S2. Data for Hydrolase Receptors

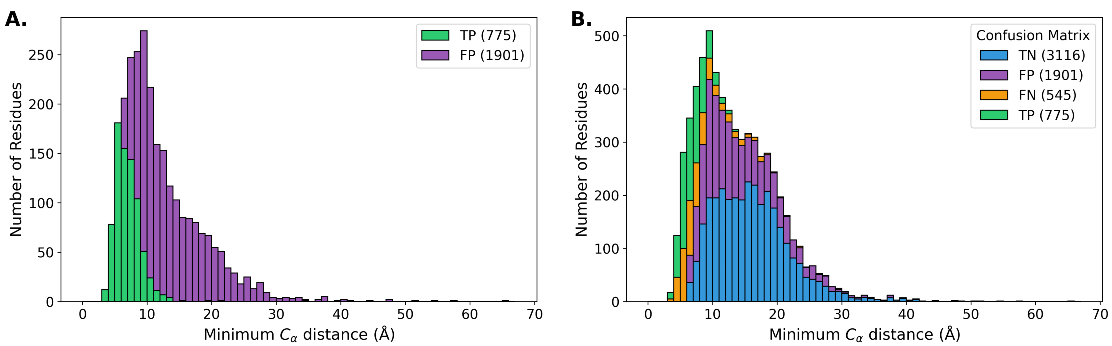
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2LBM	15001	17569	0.60	0.46		2JZI	51289	15624	0.43	0.22
	6FDP	19757	34223	0.48	0.43	7QCX	1D5G	34688	4516	0.27	0.22
2GS0	2LOX	6225	18229	0.47	0.42		6OQJ	4276	30605	0.32	0.22
	2PLD	5318	5310	0.46	0.40		2KWI	15524	15525	0.40	0.15
	6OQJ	30599	30605	0.47	0.31		1VJ6	5131	6060	0.29	0.15
	2RQG	4915	10019	0.43	0.30		1ONV	27288	5685	0.34	0.12
	6GC3	17173	34260	0.29	0.28	2KQ3	2KHS	16585	15357	0.31	0.11
	2LOZ	10318	17364	0.38	0.27		2LE8	16396	17702	0.21	0.10
	1CF4	18251	4700	0.49	0.27		2HUG	6592	7241	0.35	0.09
	2KC8	16066	16065	0.52	0.27		2LTO	10276	18490	0.14	0.09
	6R5G	28070	34384	0.39	0.25	2L5U	2L75	17285	17344	0.44	0.02
	2K7L	27288	15919	0.39	0.25		7NQC	51289	34608	0.34	0.01
	2KYL	15972	16967	0.41	0.23		2RMK	5511	11010	0.18	-0.02
	2KWI	15230	15525	0.36	0.23	2JNS	6BGG	15125	30367	0.14	-0.08



SI Fig. S1. Confusion Matrix Histograms for Hydrolase Receptors

SI Table S3. Data for Transferase Receptors

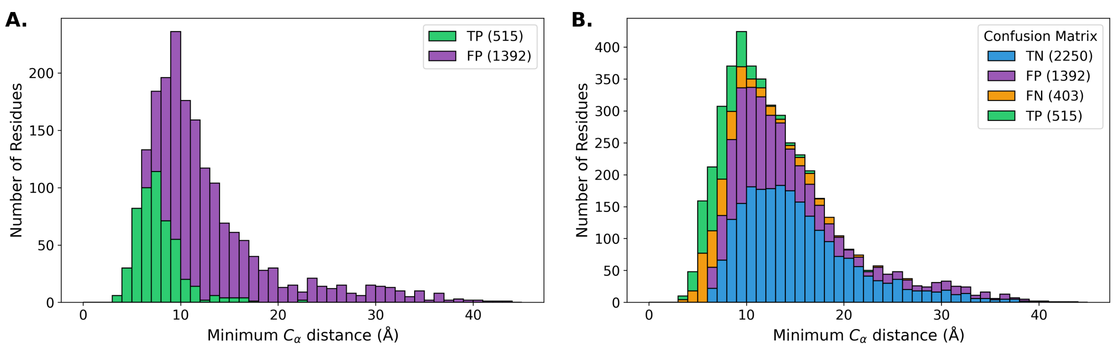
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2MRE	25071	25070	0.80	0.60
	2N9P	25913	25914	0.72	0.58
7JMY	7JYN	30782	30790	0.53	0.43
	2N8T	25866	25865	0.64	0.42
	2K7A	11018	16809	0.54	0.42
	2C52	15398	6874	0.56	0.41
	2PLD	5318	5310	0.46	0.40
5AHT	2KPZ	25349	16574	0.55	0.40
1W1F	1WA7	6261	6456	0.51	0.38
2LSY	2N1G	18455	25559	0.45	0.37
	2KWU	17769	16880	0.40	0.37
2KM4	2L0I	16411	17044	0.36	0.36
	2KWV	17769	16885	0.39	0.34
2LSG	2LSJ	18431	18433	0.35	0.34
	2MC6	19428	19429	0.47	0.33
2LSY	2LSK	18455	18434	0.38	0.32
	2LVO	18581	18582	0.50	0.30
5AHT	2KQ0	25349	16575	0.50	0.30
	2OJ2	4122	6581	0.37	0.28
	5VF0	17769	30276	0.31	0.27
	5JYV	30091	30093	0.45	0.27
	1CF4	18251	4700	0.49	0.27
2JNS	2NCZ	15125	26041	0.40	0.25
	2LGK	17813	17812	0.52	0.25
	2MOW	17173	19954	0.21	0.25
	2MBB	15410	19394	0.43	0.24
2L0I	5LVF	17044	34041	0.31	0.24
1U2N	1R8U	6268	5987	0.56	0.24
	1I5H	26698	4963	0.40	0.23
	2KYL	15972	16967	0.41	0.23
	8SG2	27579	31080	0.44	0.23
1U2N	1L8C	6268	5327	0.58	0.22
	7ZEY	34724	34727	0.36	0.22
1U2N	2KA4	6268	16014	0.55	0.21
	2MH0	34231	19610	0.33	0.21
	2MPS	6612	18876	0.34	0.20
	6U19	16620	27875	0.43	0.20
	2LVO	18610	18582	0.34	0.19
	2LAS	50128	25553	0.28	0.19
2L0I	5M9D	17044	34058	0.33	0.18
	2YS5	11095	11094	0.35	0.18
6VED	2L3R	30703	17200	0.30	0.18
	5J8H	51289	30063	0.43	0.18
	2N83	25828	25833	0.36	0.17
	5IAY	30703	30019	0.22	0.17
	2LXS	6095	18694	0.32	0.17
	2MV7	18094	19516	0.38	0.15
	7ZEY	16983	34727	0.44	0.14
	2MRE	18610	25070	0.28	0.13
	5JYV	30092	30093	0.31	0.13
	9C5E	18610	31177	0.32	0.13
	6QXZ	27250	27251	0.42	0.13
	2H7D	15792	7150	0.24	0.11
	2M86	15945	19230	0.25	0.10
	2N1A	6304	25553	0.10	0.10
	2RSE	50325	11471	0.10	0.10
	2KA6	4789	16015	0.51	0.09
	2LTO	10276	18490	0.14	0.09
	2K7A	5461	16809	0.21	0.09
	2KJE	4789	16318	0.45	0.07
	2MZD	34231	25484	0.35	0.06
	2RSY	7141	11508	0.30	0.05
	2G35	15792	7061	0.17	0.04
	2MSR	34179	25130	0.12	0.00
	2RMK	5511	11010	0.18	-0.02
	2LGG	17813	17808	0.37	-0.03
	2I94	5332	7293	0.17	-0.03
	2K8F	34231	15944	0.22	-0.05
2JNS	2ND1	15125	26043	0.23	-0.06
	2LV6	51289	18556	0.35	-0.07
	2N83	25883	25833	0.29	-0.10
	9C5E	18642	31177	0.35	-0.11
	2RSE	16931	11471	0.00	-0.16
	2JMF	6262	15016	0.13	-0.17



SI Fig. S2. Confusion Matrix Histograms for Transferase Receptors

SI Table S4. Data for All Alpha Receptors

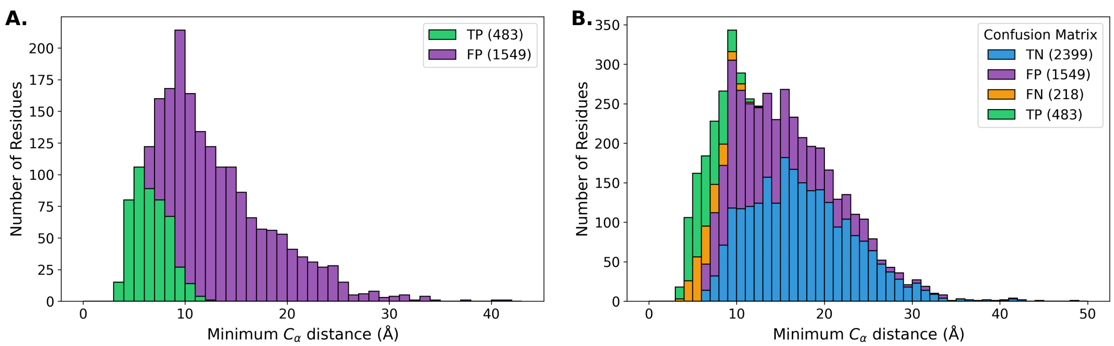
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	6FDP	19757	34223	0.48	0.43		2LXS	6095	18694	0.32	0.17
	2M5A	6804	19043	0.62	0.42		1LXF	30966	5386	0.37	0.14
	2C52	15398	6874	0.56	0.41		2M0K	51289	17771	0.41	0.14
	2L6E	16555	17307	0.52	0.40		2KTB	51472	16694	0.20	0.13
	6FDT	19757	34224	0.33	0.34		1PD7	6899	5808	0.37	0.13
	2M5A	5656	19043	0.53	0.34		2KNE	51289	16465	0.36	0.13
	2JXC	4184	15554	0.58	0.33		1ONV	27288	5685	0.34	0.12
	2KSP	15279	16671	0.40	0.32		2MES	51289	19238	0.45	0.11
	2RQG	4915	10019	0.43	0.30		2MGU	51289	19604	0.42	0.09
	2LY4	11532	18709	0.40	0.30		2L53	51289	17264	0.27	0.06
	2MG5	51289	19586	0.44	0.27		1SY9	51289	5480	0.36	0.06
	2LP0	25142	17407	0.49	0.26		6BUT	51289	27095	0.25	0.04
	1H8B	17627	4453	0.33	0.25		2M56	4154	19038	0.12	0.04
	2KNH	7396	16467	0.30	0.24		2M0J	51289	17771	0.35	0.03
	2JZI	51289	15624	0.43	0.22		2MSR	34179	25130	0.12	0.00
	2LL6	51289	18027	0.39	0.21		2N80	25883	25829	0.26	-0.02
	2MPS	6612	18876	0.34	0.20		2I94	5332	7293	0.17	-0.03
	2M56	17415	19038	0.12	0.18		5TP6	51289	30196	0.21	-0.03
	5J8H	51289	30063	0.43	0.18		1NPQ	4232	5738	0.24	-0.03
	2N83	25828	25833	0.36	0.17		2N83	25883	25833	0.29	-0.10
	2LL7	51289	18028	0.33	0.17		2N80	51835	25829	0.04	-0.13



SI Fig. S3. Confusion Matrix Histograms for All Alpha Receptors

SI Table S5. Data for All Beta Receptors

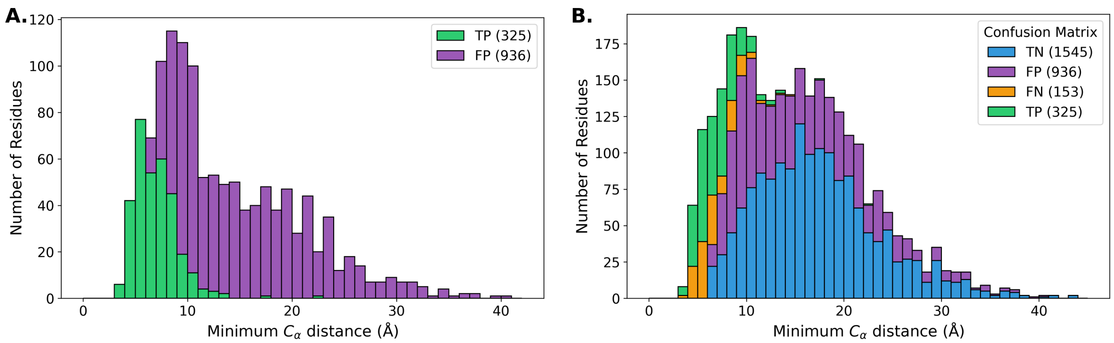
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	1YWI	6658	6659	0.78	0.72
	2MCN	16641	19447	0.68	0.58
	2FIN	5073	7024	0.65	0.52
	2ROZ	10235	10236	0.59	0.50
	5I22	4871	30010	0.55	0.49
	2AFF	5959	6748	0.62	0.48
2RQT	2RQU	11081	11082	0.60	0.42
2GS0	2LOX	6225	18229	0.47	0.42
	2K7A	11018	16809	0.54	0.42
	1N7T	18785	5631	0.44	0.40
1W1F	1WA7	6261	6456	0.51	0.38
2GS0	2MKR	6225	19791	0.29	0.33
2GS0	2N0Y	6225	25540	0.37	0.33
	2M0U	18824	18825	0.35	0.31
	2RUK	4901	11578	0.44	0.30
	2MS4	2208	25104	0.29	0.29
	2KPL	6911	16559	0.40	0.28
	2LOZ	10318	17364	0.38	0.27
	2L29	17128	17127	0.26	0.26
2GS0	2M14	6225	18842	0.24	0.23
	1I5H	26698	4963	0.40	0.23
	2L4T	17254	17255	0.31	0.23
2GS0	5URN	6225	30243	0.29	0.23
	2M3M	17373	17942	0.42	0.22
	1DDM	4683	4263	0.22	0.22
7QCX	1D5G	34688	4516	0.27	0.22
2GS0	2N23	6225	25584	0.17	0.22
1SSF	2LVM	5878	18579	0.31	0.20
	2RVB	4901	11594	0.44	0.20
	2KVQ	15490	16788	0.25	0.16
	2FFK	6809	7024	0.16	0.15
	1VJ6	5131	6060	0.29	0.15
	2NMB	4683	4263	0.15	0.13
2KQ3	2KHS	16585	15357	0.31	0.11
	2MCN	18610	19447	0.40	0.10
	2KOH	27205	16507	0.30	0.10
	2FIN	6809	7024	0.10	0.09
	5XV8	4901	36101	0.37	0.09
	2HUG	6592	7241	0.35	0.09
	2K7A	5461	16809	0.21	0.09
	2LZ6	18610	18737	0.29	0.08
	2G35	15792	7061	0.17	0.04
	2LZ6	15407	18737	0.28	0.02
	2KVM	15674	16778	0.33	-0.00
	2N80	25883	25829	0.26	-0.02
	2MNU	4283	19906	0.19	-0.04
	2L29	34000	17127	0.16	-0.08
	2N80	51835	25829	0.04	-0.13
	2JMF	6262	15016	0.13	-0.17



SI Fig. S4. Confusion Matrix Histograms for All Beta Receptors

SI Table S6. Data for Alpha and Beta (a+b) Receptors

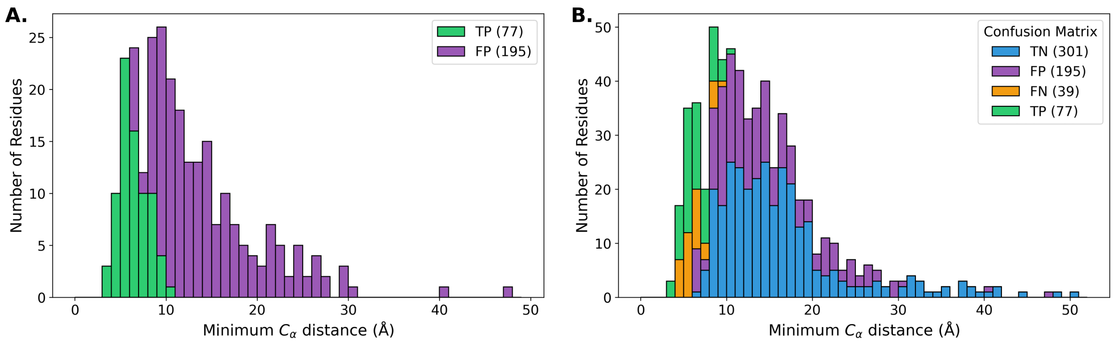
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2MRE	25071	25070	0.80	0.60		2MBB	15410	19394	0.43	0.24
	2MCN	16641	19447	0.68	0.58		1KLQ	52275	5299	0.32	0.22
	2FIN	5073	7024	0.65	0.52		2LVO	18610	18582	0.34	0.19
	2K7A	11018	16809	0.54	0.42		2M56	17415	19038	0.12	0.18
	2PLD	5318	5310	0.46	0.40		6H8C	18827	34307	0.29	0.17
2LD9	2MBH	15410	19399	0.40	0.40		2MRE	18610	25070	0.28	0.13
	2KWU	17769	16880	0.40	0.37		2MCN	18610	19447	0.40	0.10
	2LI5	16835	17879	0.42	0.36		2N1A	6304	25553	0.10	0.10
	2G3Q	4769	7002	0.38	0.35		2FIN	6809	7024	0.10	0.09
	2KWV	17769	16885	0.39	0.34		2K7A	5461	16809	0.21	0.09
	1M4P	50765	5532	0.29	0.32		6XMN	30316	50331	0.40	0.09
	2MUR	17769	25230	0.40	0.32		2LZ6	18610	18737	0.29	0.08
	2LVO	18581	18582	0.50	0.30		2N55	52209	25694	0.44	0.07
	5VF0	17769	30276	0.31	0.27		2RSY	7141	11508	0.30	0.05
	2M0G	19034	18808	0.43	0.27		2M56	4154	19038	0.12	0.04
	2MUR	25229	25230	0.57	0.26		2LZ6	15407	18737	0.28	0.02
	2K6Q	5958	15877	0.36	0.26		2K6D	52080	15866	0.15	-0.18
	2M0G	18802	18808	0.29	0.25						



SI Fig. S5. Confusion Matrix Histograms for Alpha and Beta (a+b) Receptors

SI Table S7. Data for BET-ET Domain Receptors

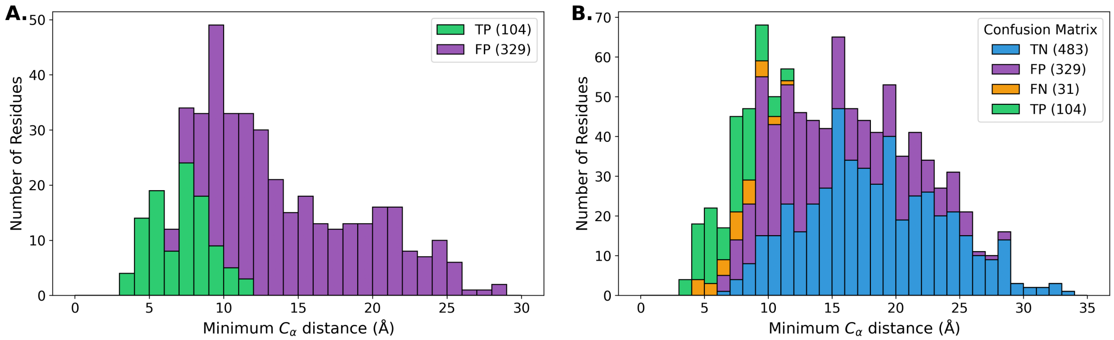
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
7JMY	7JYN	30782	30790	0.53	0.43	2JNS	2NCZ	15125	26041	0.40	0.25
7JMY	7JYZ	30782	30791	0.50	0.40	2JNS	2ND0	15125	26042	0.31	0.09
7JMY	7JQ8	30782	30786	0.51	0.37	2JNS	2ND1	15125	26043	0.23	-0.06
2JNS	6BNH	15125	30373	0.48	0.36	2JNS	6BGG	15125	30367	0.14	-0.08



SI Fig. S6. Confusion Matrix Histograms for BET-ET Domain Receptors

SI Table S8. Data for TFIIH Domain Receptors

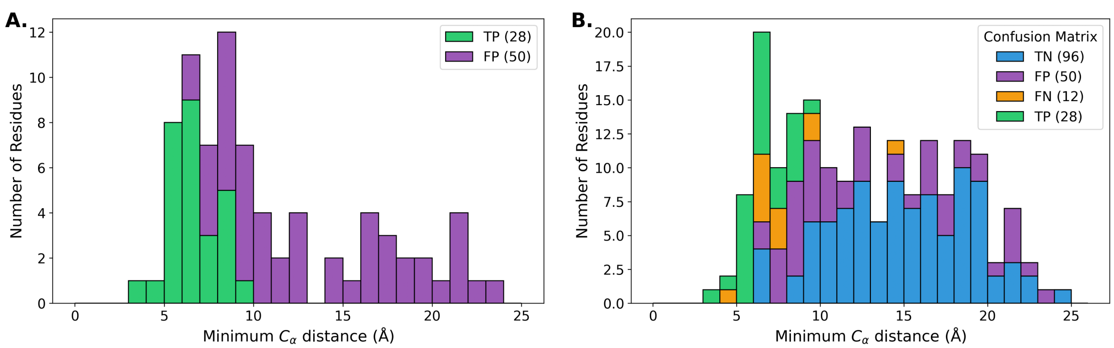
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
2GS0	2LOX	6225	18229	0.47	0.42	2GS0	5URN	6225	30243	0.29	0.23
2GS0	2MKR	6225	19791	0.29	0.33	2GS0	2N23	6225	25584	0.17	0.22
2GS0	2N0Y	6225	25540	0.37	0.33		2RVB	4901	11594	0.44	0.20
	2RUK	4901	11578	0.44	0.30		5XV8	4901	36101	0.37	0.09
2GS0	2M14	6225	18842	0.24	0.23						



SI Fig. S7. Confusion Matrix Histograms for TFIIH Domain Receptors

SI Table S9. Data for Ubiquitin Domain Receptors

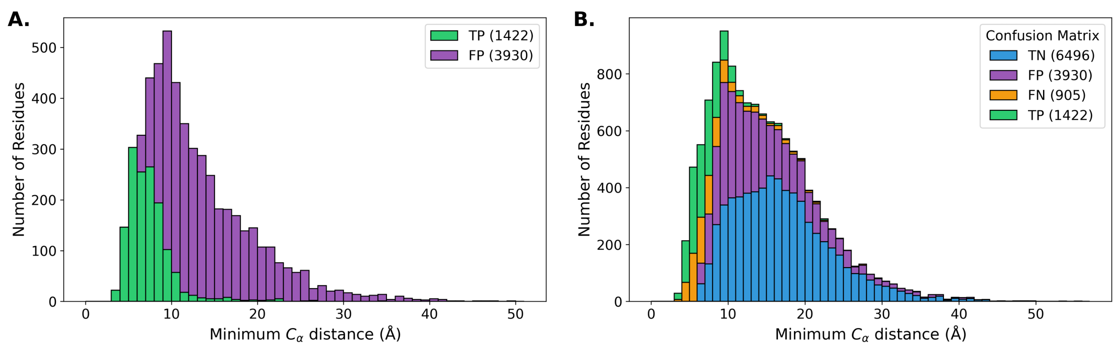
apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC	apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2MUR	17769	25230	0.40	0.32		2MUR	25229	25230	0.57	0.26
	5VF0	17769	30276	0.31	0.27						



SI Fig. S8. Confusion Matrix Histograms for Ubiquitin Domain Receptors

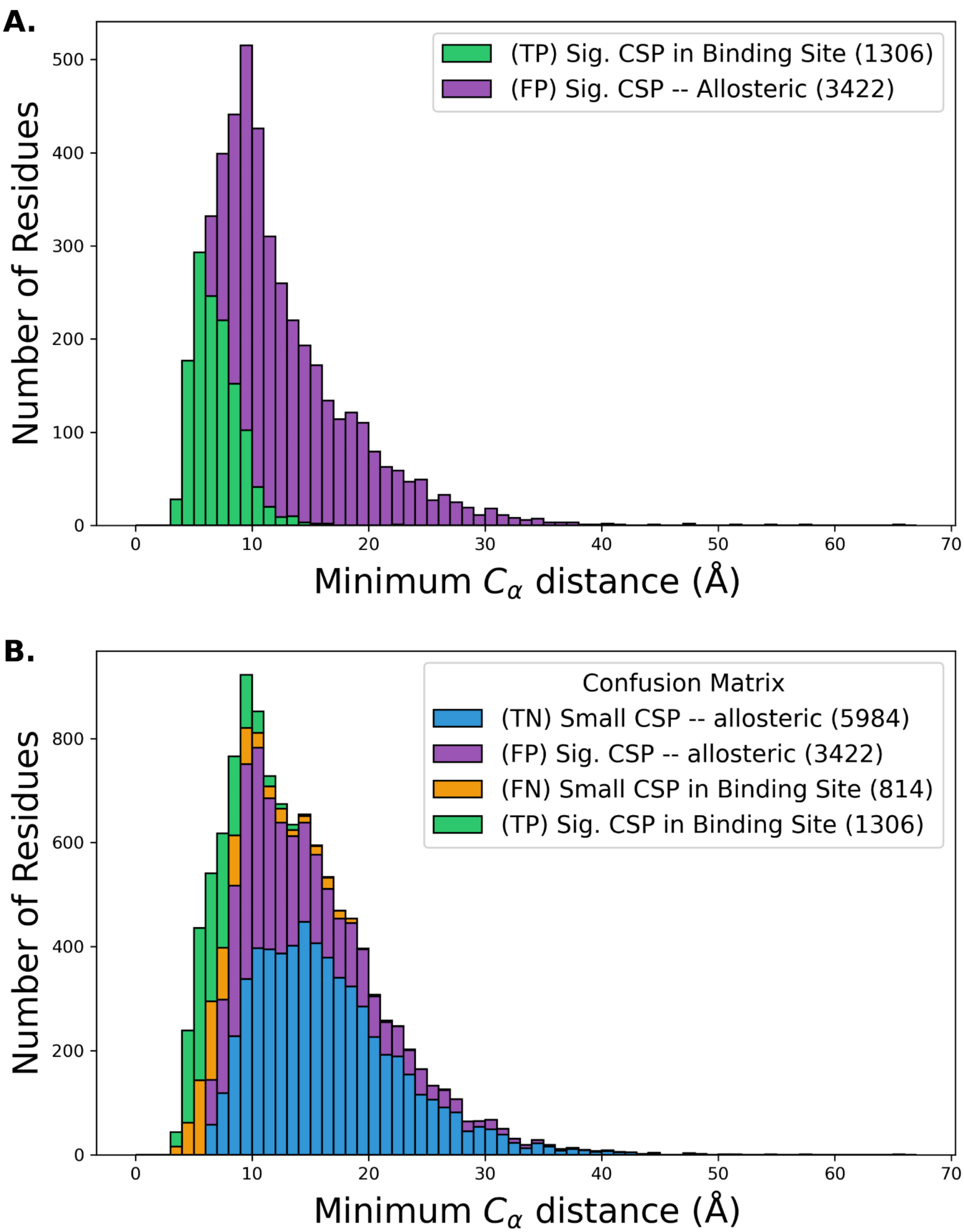
SI Table S10. Data for targets with dissimilar apo/holo experimental conditions

apo_pdb	holo_pdb	apo_bmrb	holo_bmrb	F1	MCC
	2MRE	25071	25070	0.80	0.60
	2MCN	16641	19447	0.68	0.58
	2N9P	25913	25914	0.72	0.58
	2AFF	5959	6748	0.62	0.48
	6IVU	36220	36221	0.67	0.48
1XWH	2KE1	6374	16510	0.68	0.46
	6FDP	19757	34223	0.48	0.43
2RQT	2RQU	11081	11082	0.60	0.42
2GS0	2LOX	6225	18229	0.47	0.42
	7OVC	6546	34638	0.49	0.42
	2K7A	11018	16809	0.54	0.42
	2M5A	6804	19043	0.62	0.42
5AHT	2KPZ	25349	16574	0.55	0.40
	2KID	4879	16270	0.39	0.40
2LD9	2MBH	15410	19399	0.40	0.40
	1N7T	18785	5631	0.44	0.40
	6RH6	34394	34395	0.34	0.38
	2YKA	16697	17693	0.36	0.37
	2KWU	17769	16880	0.40	0.37
2KM4	2L0I	16411	17044	0.36	0.36
	2G3Q	4769	7002	0.38	0.35
	2KWV	17769	16885	0.39	0.34
2LSG	2LSJ	18431	18433	0.35	0.34
2AWT	2N9E	6801	25901	0.30	0.33
	2JXC	4184	15554	0.58	0.33
2GS0	2MKR	6225	19791	0.29	0.33
2GS0	2N0Y	6225	25540	0.37	0.33
	2KSP	15279	16671	0.40	0.32
2LPC	2M04	18250	18793	0.45	0.31
	2BN5	6691	6690	0.28	0.31
	2KGI	16209	16210	0.62	0.30
5AHT	2KQ0	25349	16575	0.50	0.30
	2RUK	4901	11578	0.44	0.30
	2RQG	4915	10019	0.43	0.30
	2MS4	2208	25104	0.29	0.29
	2KPL	6911	16559	0.40	0.28
	2MPM	4155	19989	0.41	0.28
	2MG5	51289	19586	0.44	0.27
	5JYV	30091	30093	0.45	0.27
	2M0G	19034	18808	0.43	0.27
5YDY	2LAW	36115	17538	0.50	0.27
	1CF4	18251	4700	0.49	0.27
	2N3K	15125	25649	0.39	0.26
	2M0G	18802	18808	0.29	0.25
	2MOW	17173	19954	0.21	0.25
	2MBB	15410	19394	0.43	0.24
	2KNH	7396	16467	0.30	0.24
	2N3A	34537	25639	0.37	0.24
2L0I	5LVF	17044	34041	0.31	0.24
2GS0	2M14	6225	18842	0.24	0.23
	1I5H	26698	4963	0.40	0.23
	2IPA	6075	15028	0.18	0.23
	2KWI	15230	15525	0.36	0.23
	2MZW	25368	25504	0.29	0.23
2GS0	5URN	6225	30243	0.29	0.23
1U2N	1L8C	6268	5327	0.58	0.22
2GS0	2N23	6225	25584	0.17	0.22
	6OQJ	4276	30605	0.32	0.22
	2LL6	51289	18027	0.39	0.21
1U2N	2KA4	6268	16014	0.55	0.21
	6FGP	16803	34232	0.38	0.21
	7L8V	26682	27692	0.29	0.21
	2MH0	34231	19610	0.33	0.21
	2MPS	6612	18876	0.34	0.20
	2RVB	4901	11594	0.44	0.20
	6U19	16620	27875	0.43	0.20
	6CO4	30824	25979	0.31	0.20
	2LVO	18610	18582	0.34	0.19
	2LAS	50128	25553	0.28	0.19
2L0I	5M9D	17044	34058	0.33	0.18
	2MMA	19849	19850	0.30	0.18
	2N83	25828	25833	0.36	0.17
	2LL7	51289	18028	0.33	0.17
	5IAY	30703	30019	0.22	0.17
	5J7J	51289	30062	0.48	0.17
	2LXS	6095	18694	0.32	0.17
2M03	6IJQ	18792	36133	0.18	0.16
	2FFK	6809	7024	0.16	0.15
1N3H	2L1C	5566	17080	0.34	0.15
	1LXF	30966	5386	0.37	0.14
2JQ6	2KFH	15279	16181	0.27	0.14
	9C5E	18610	31177	0.32	0.13
	2KNE	51289	16465	0.36	0.13
	6QXZ	27250	27251	0.42	0.13
	1ONV	27288	5685	0.34	0.12
2KJD	2M0V	16637	18826	0.20	0.11
1V49	2LUE	5958	18518	0.22	0.11
	2M86	15945	19230	0.25	0.10
	2MCN	18610	19447	0.40	0.10
	2N1A	6304	25553	0.10	0.10
	5XV8	4901	36101	0.37	0.09
	2KA6	4789	16015	0.51	0.09
	6XMN	30316	50331	0.40	0.09
	2MGU	51289	19604	0.42	0.09
	2KJE	4789	16318	0.45	0.07
	2L53	51289	17264	0.27	0.06
	2MZD	34231	25484	0.35	0.06
	6BUT	51289	27095	0.25	0.04
	2MZP	30966	25495	0.17	0.04
	9R3Y	26643	34992	0.25	0.04
	2MKP	25495	19789	0.27	0.04
	2LZ6	15407	18737	0.28	0.02
	7NQC	51289	34608	0.34	0.01
	7F7X	17769	36427	0.20	0.00
	6B1G	27072	30345	0.26	-0.01
2E5E	2L7U	7364	17378	0.14	-0.01
5YDX	2LAY	36114	17540	0.29	-0.01
	2JMX	6564	15072	0.09	-0.01
	2N8J	51289	25852	0.19	-0.01
	2M55	51289	19036	0.24	-0.01
	2RMK	5511	11010	0.18	-0.02
	2NBV	30824	25995	0.15	-0.02
	5TP6	51289	30196	0.21	-0.03
	2MNU	4283	19906	0.19	-0.04
	2LV6	51289	18556	0.35	-0.07
	2L29	34000	17127	0.16	-0.08
	9C5E	18642	31177	0.35	-0.11
	2N80	51835	25829	0.04	-0.13
	2RSE	16931	11471	0.00	-0.16
	2RS9	19125	11463	0.00	-0.17

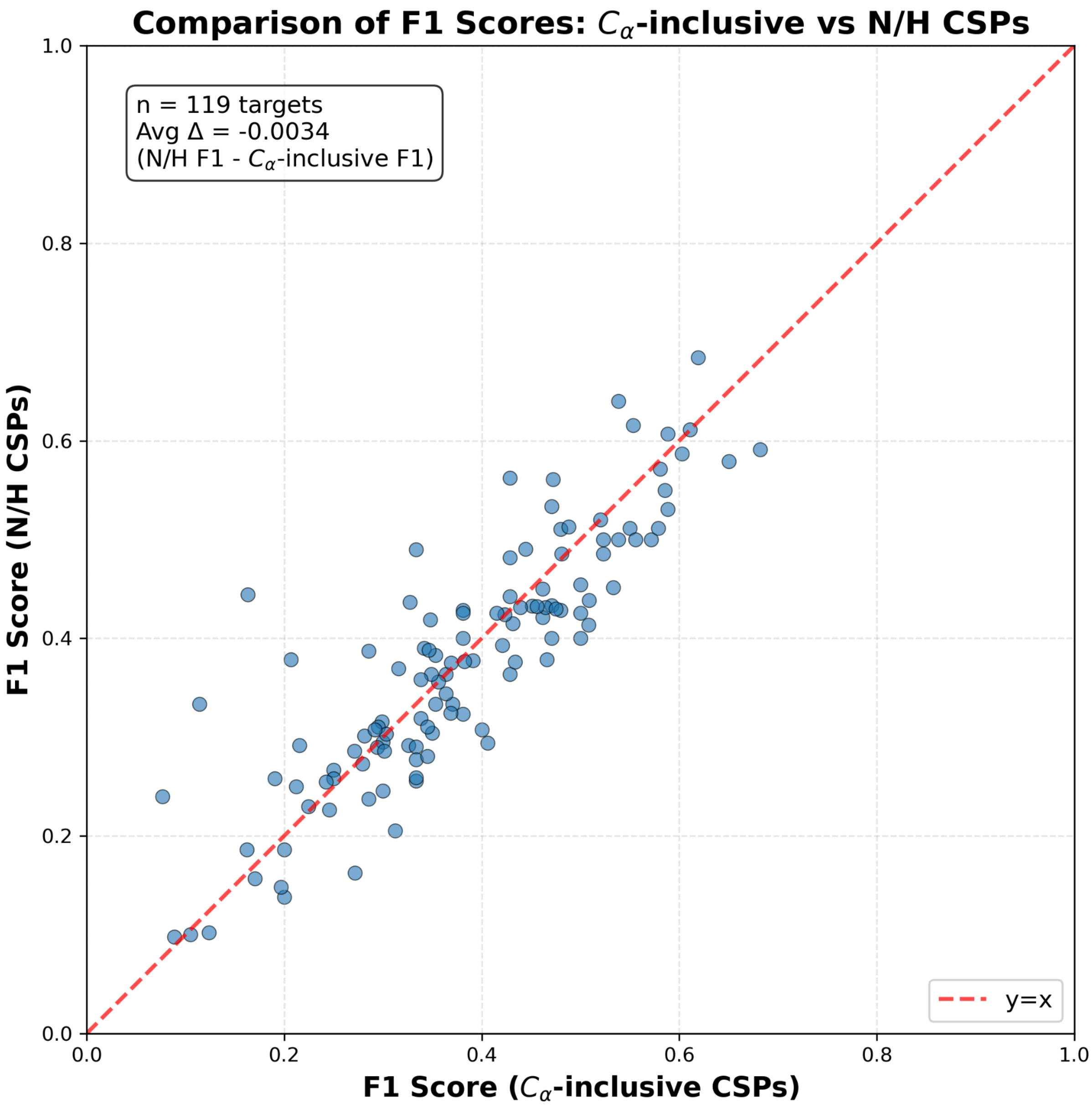


SI Fig. S9. Confusion Matrix Histograms for targets with dissimilar apo/holo experimental conditions

A.

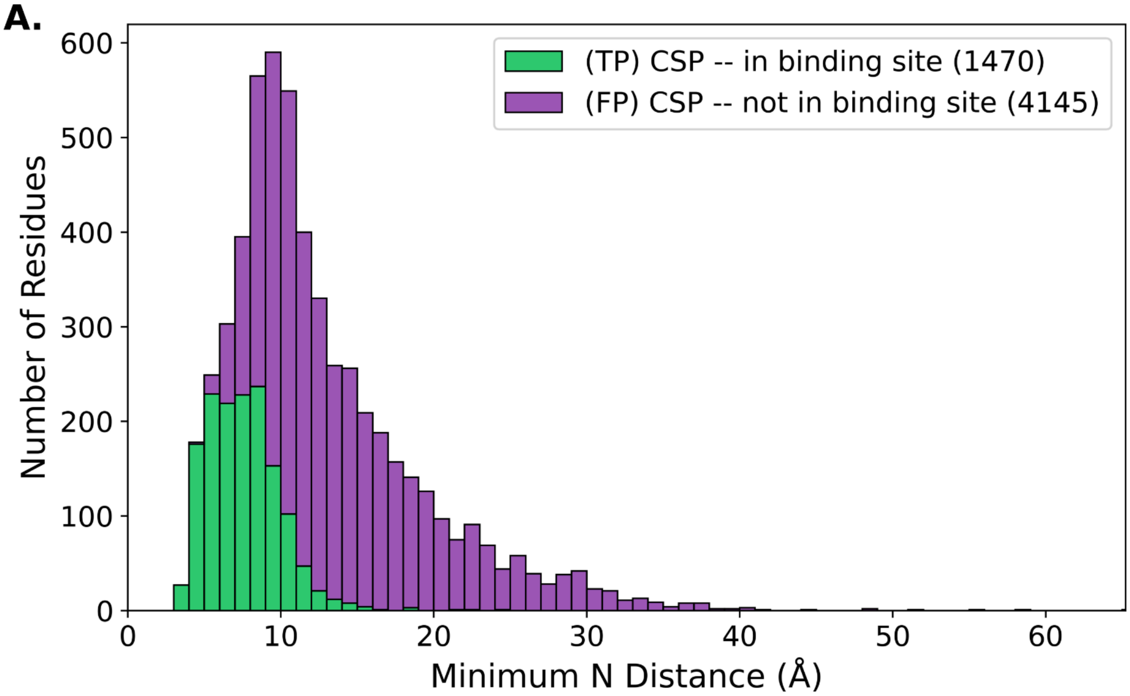


SI Fig. S10. **CA-inclusive CSP Confusion Matrix Histograms**. Calculating CSPs with CAs included in the formula does drastically not change the percentage of allosteric CSPs ~72%. CSPs are calculated using formula described in **Eqn. 2**.

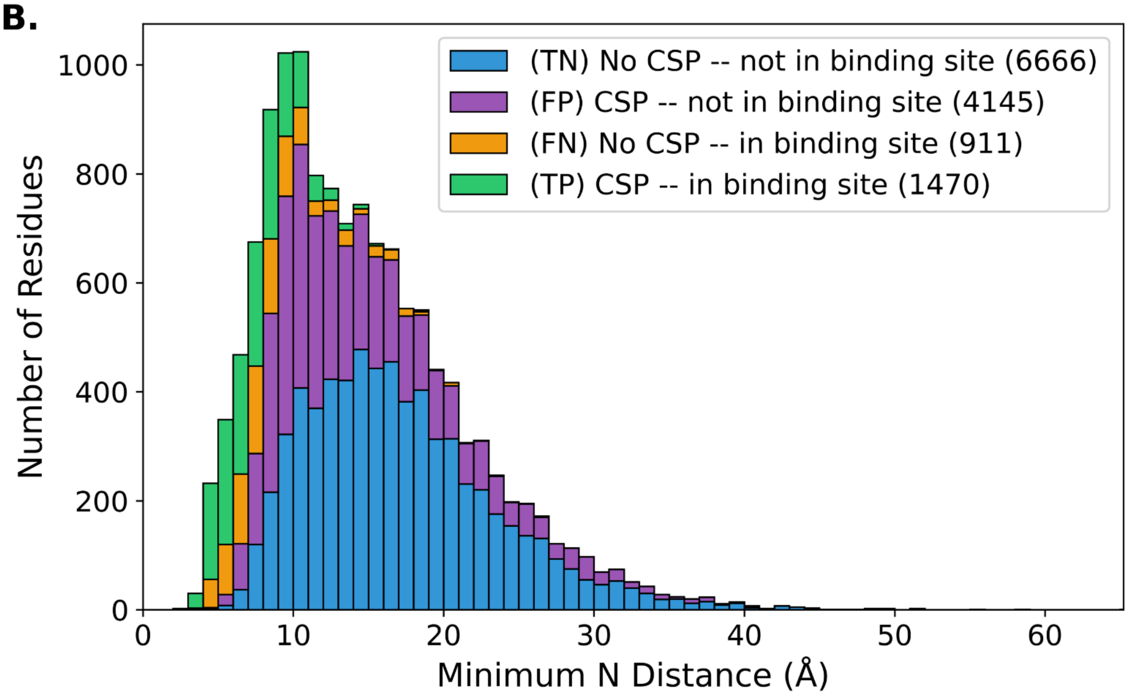


SI Fig. S11. Change in F1 scores when considering CA-inclusive CSPs. This plot shows that over the targets in the CSP DB, the average change in F1 score is negligible when comparing CSPs calculated from N/H shifts vs CSPs calculated with N/H/Ca shifts. CA-inclusive CSPs are calculated following **Eqn. 2**.

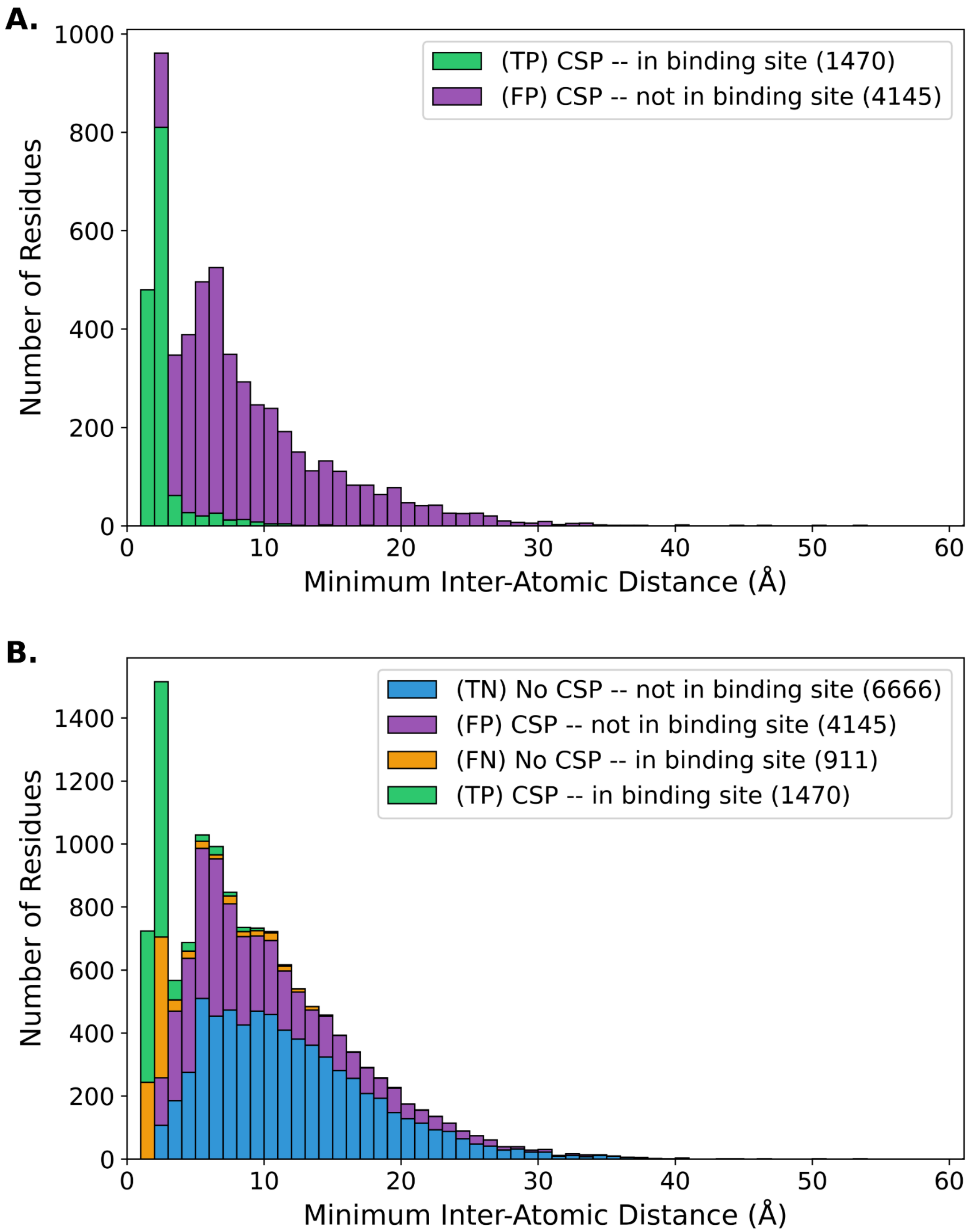
A.



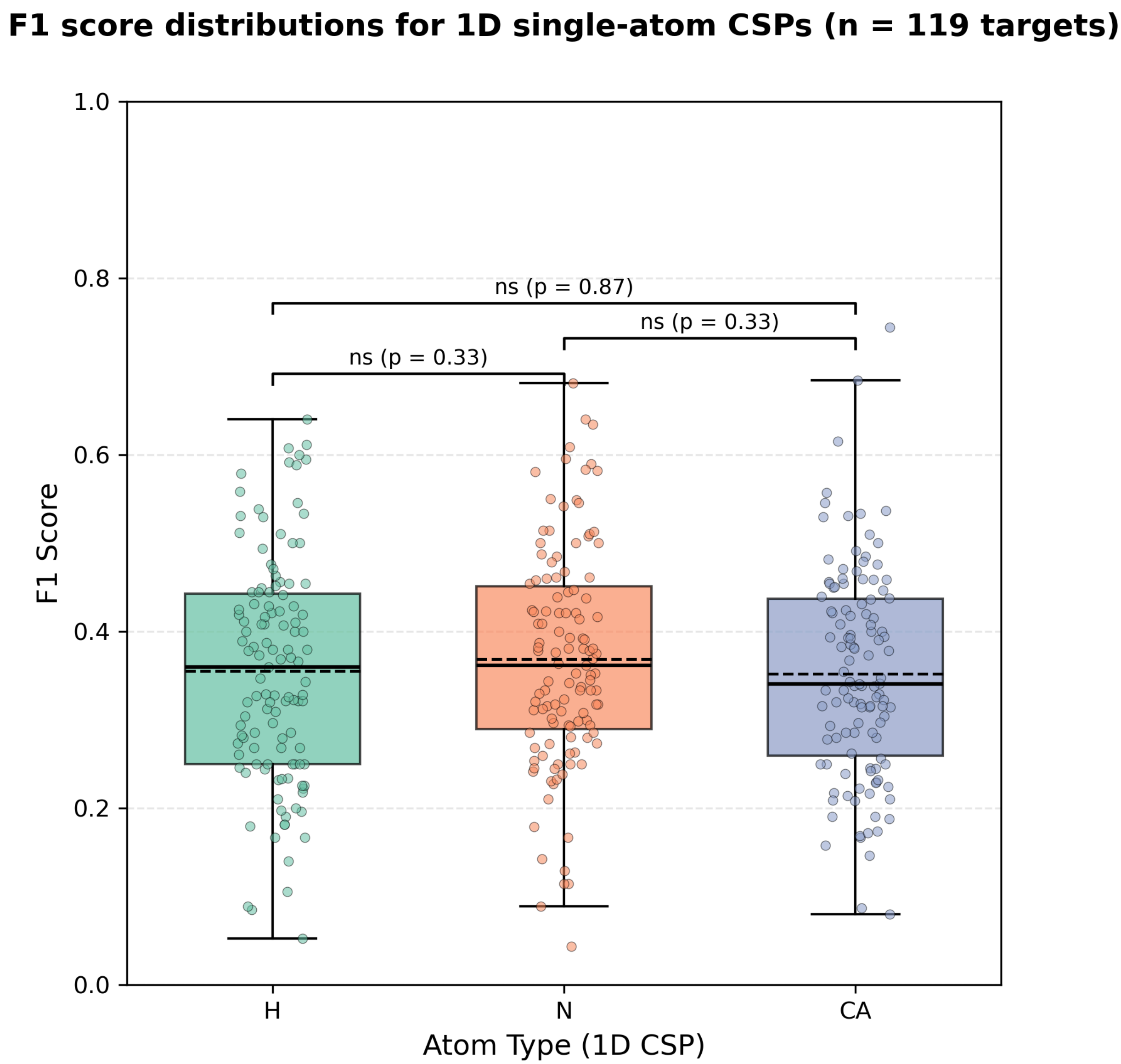
B.



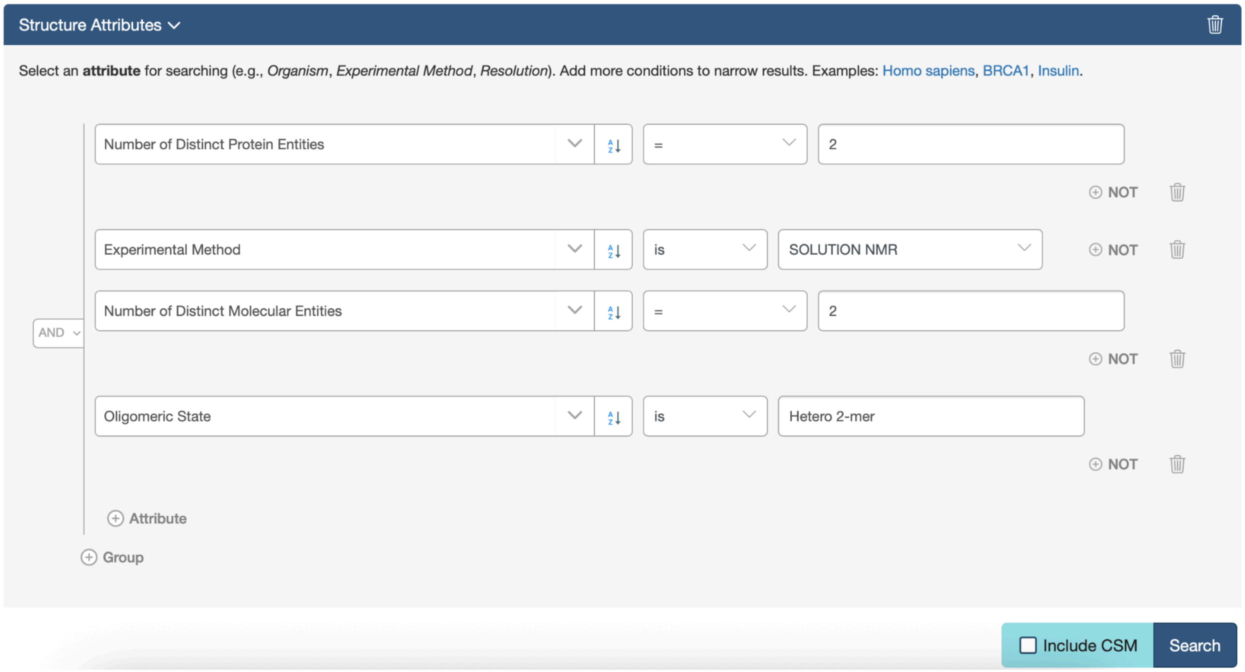
SI Fig. S12. CSP DB Confusion Matrix Histograms by closest interchain N-N distance.



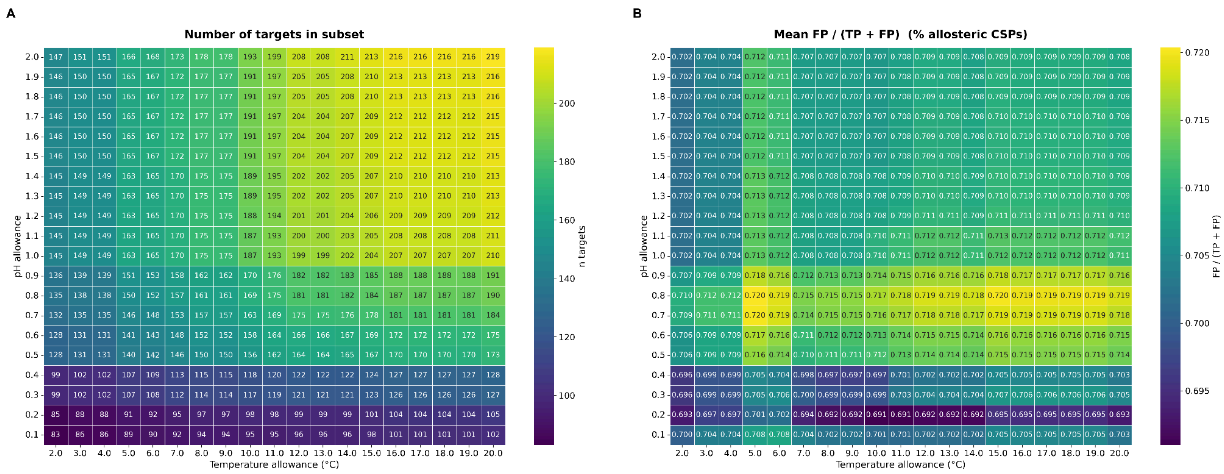
SI Fig. S13. CSP DB Confusion Matrix Histograms by closest interchain atom-atom distance.



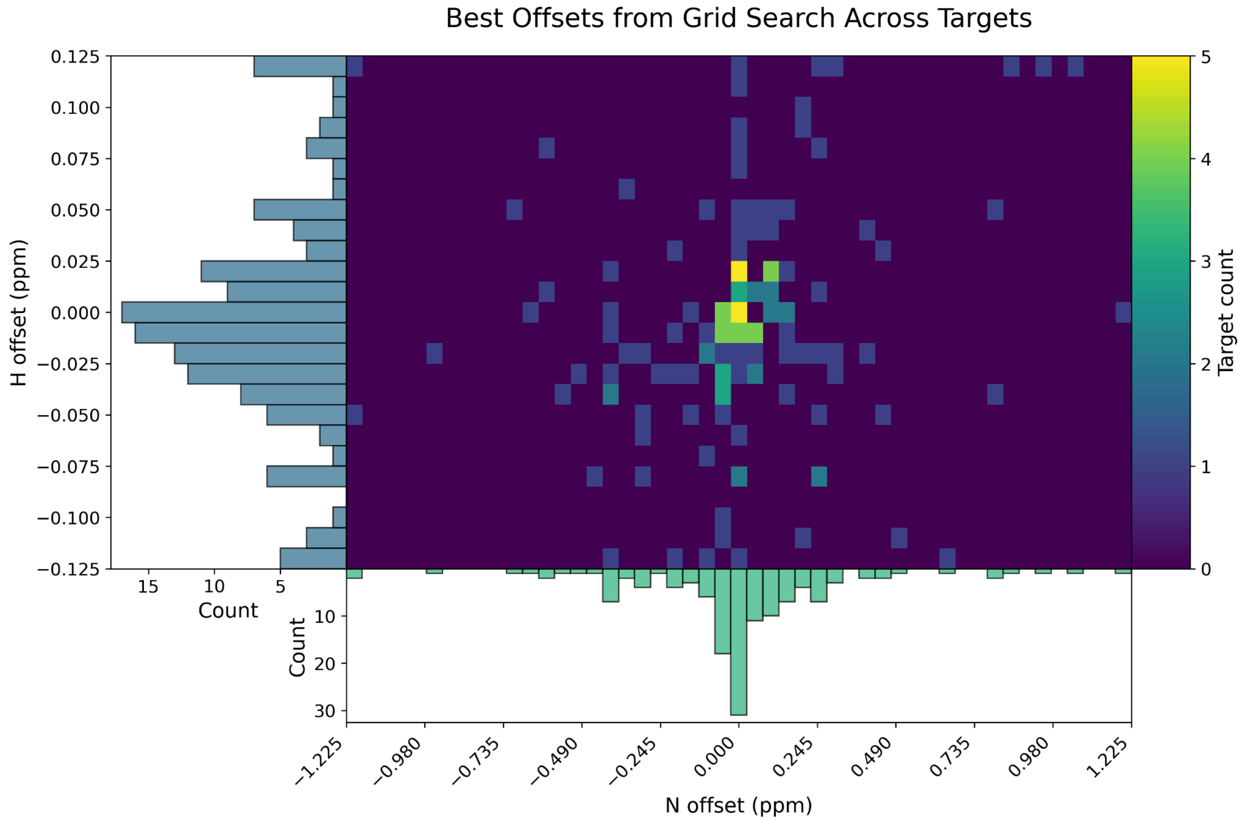
SI Fig. S14. F1 Scores for 1D CSPs. This plot shows there are no statistical differences between distributions of calculated F1 scores from 1D CSPs H/N/CA atom types. CSPs are calculated as root-squared differences after applying an ideal holo offset for a 3D H/N/CA offset grid (see **Methods** for details). Statistical analysis is performed using paired Wilcoxon tests with Holm correction.



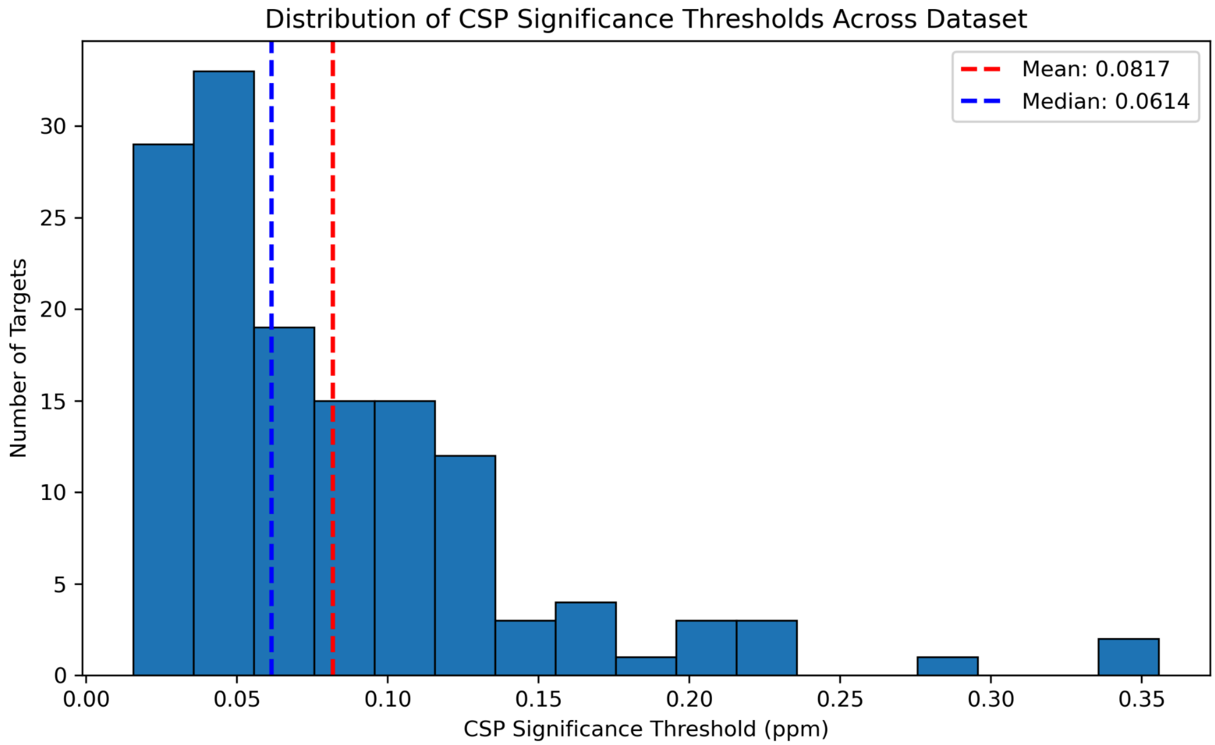
SI Fig. S15. PDB Advanced Search Settings. For the subset of targets manually scraped from the PDB, we restrict our analysis to these settings: entries resolved by Solution NMR, with two and only two distinct protein chains. At the time of curation, this advanced search resulted in 720 hits on the RCSB PDB. From this list of 720, entries were included in the CSP DB if there were N and H chemical shifts deposited in the BMRB for an identical sequence which was evaluated in both apo and holo experiments.



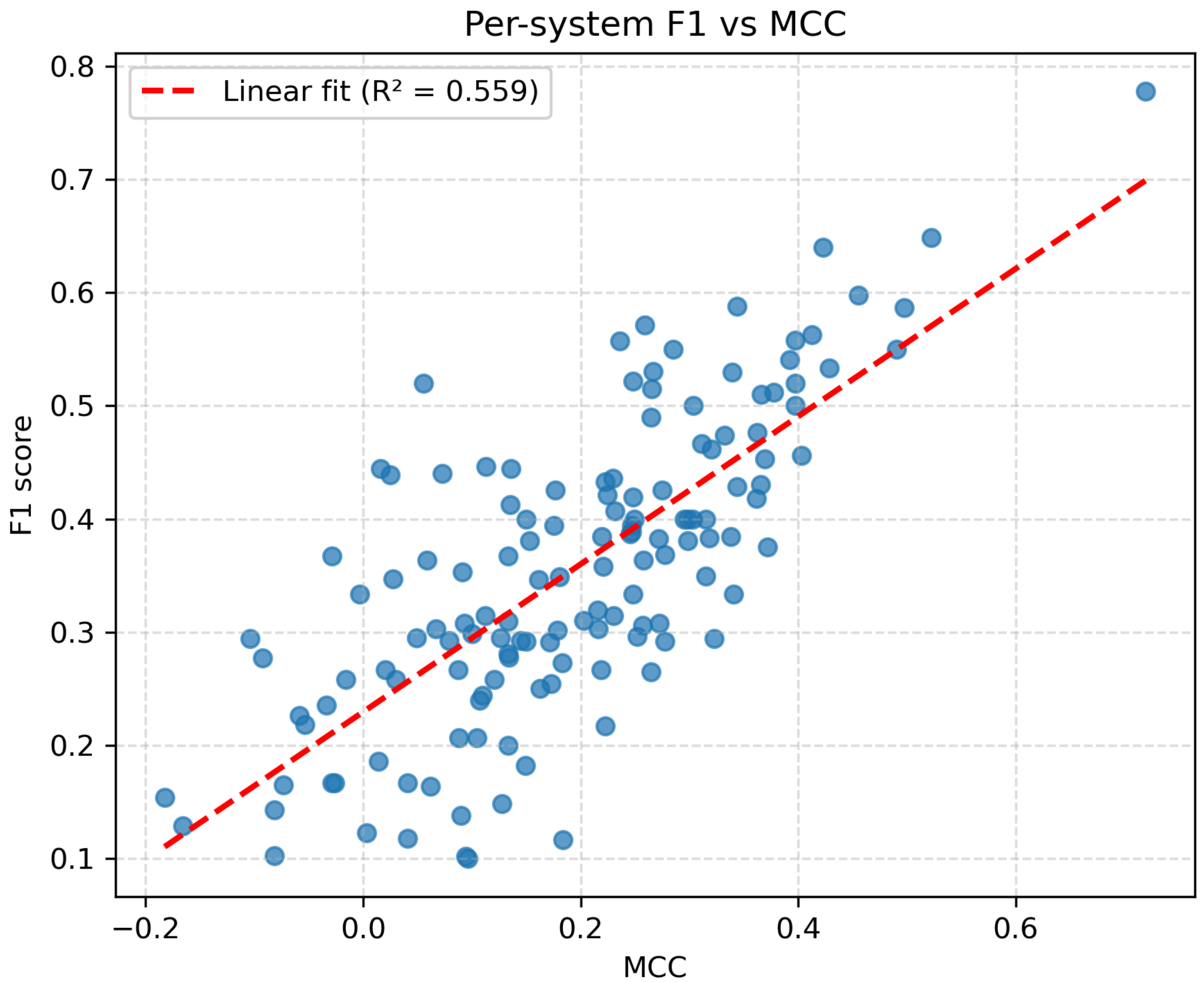
SI Fig. S16. Impact of Experiment Condition Thresholding on Results. These heatmaps demonstrate that changing the threshold of difference in pH and temperature between the apo and holo HSQC experiments does not have a strong effect on the reported outcomes of this study. 219 receptors have experimental condition differences within ± 2 pH units and ± 20 ºC, and the mean percentage of allosteric CSPs across this more lenient thresholding is only slightly less (70.6%) than the value reported in the main text on the subset with conditions ± pH 0.5 units and ± 5 ºC (71.6%).



SI Fig. S17. **Ideal offsets across the whole dataset for N/H**. This heatmap shows the offsets used to align the apo/holo HSQC plots. The majority of targets require small adjustments within 0.025 ppm on the Hydrogen axis and 0.245 ppm on the Nitrogen axis.



SI Fig. S18. CSP significance threshold Histogram. This histogram summarizes the thresholds for determining whether a CSP for a given receptor is significant. For a small number of targets, the calculated significance threshold is significantly higher than one would expect, we attribute this to differences in experimental conditions between the apo and holo NMR experiments which aberrantly perturbs all observed shifts. We prune from the CSPdb any targets with a calculated significance threshold > 0.6 ppm. This includes the following targets: 2LMC, 1OO9, 2L12, 2L00, 2N01, 2MP0, 8VOI, 8U2Y, 6OSW.



SI Fig. S19. F1 vs MCC score comparison across CSP DB targets. This plot shows the correlation between calculated F1 and MCC metrics. MCC is a more reliable metric for larger systems with many True Negatives (small CSPs outside of the binding site) (7).

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